

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("smartcart_customers.csv")
```

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome
0	5524	1957	Graduation	Single	58138.0	0
1	2174	1954	Graduation	Single	46344.0	1
2	4141	1965	Graduation	Together	71613.0	0
3	6182	1984	Graduation	Together	26646.0	1
4	5324	1981	PhD	Married	58293.0	1

	Dt_Customer	Recency	MntWines	...	MntFishProducts
0	04-09-2012	58	635	...	172
1	08-03-2014	38	11	...	2
2	21-08-2013	26	426	...	111
3	10-02-2014	26	11	...	10
4	19-01-2014	94	173	...	46

	MntGoldProds	NumDealsPurchases	NumWebPurchases
0	88	3	8
1	6	2	1
2	42	1	8
3	5	2	2
4	15	5	5

	NumStorePurchases	NumWebVisitsMonth	Complain	Response
0	4	7	0	1
1	2	5	0	0

2	10	4	0	0
3	4	6	0	0
4	6	5	0	0

[5 rows x 22 columns]

df.shape

(2240, 22)

df.isnull().sum()

ID	0
Year_Birth	0
Education	0
Marital_Status	0
Income	24
Kidhome	0
Teenhome	0
Dt_Customer	0
Recency	0
MntWines	0
MntFruits	0
MntMeatProducts	0
MntFishProducts	0
MntSweetProducts	0
MntGoldProds	0
NumDealsPurchases	0
NumWebPurchases	0
NumCatalogPurchases	0
NumStorePurchases	0
NumWebVisitsMonth	0
Complain	0
Response	0
dtype: int64	

Data preprocessing

1. Handle Missing Values

```
df["Income"]=df["Income"].fillna(df["Income"].median())
```

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome
Teenhome	\					
0	5524	1957	Graduation	Single	58138.0	0
0						
1	2174	1954	Graduation	Single	46344.0	1
1						

2	4141	1965	Graduation	Together	71613.0	0
0						
3	6182	1984	Graduation	Together	26646.0	1
0						
4	5324	1981	PhD	Married	58293.0	1
0						

	Dt_Customer	Recency	MntWines	...	MntFishProducts
	MntSweetProducts \				
0	04-09-2012	58	635	...	172
88					
1	08-03-2014	38	11	...	2
1					
2	21-08-2013	26	426	...	111
21					
3	10-02-2014	26	11	...	10
3					
4	19-01-2014	94	173	...	46
27					

	MntGoldProds	NumDealsPurchases	NumWebPurchases
	NumCatalogPurchases \		
0	88	3	8
10			
1	6	2	1
1			
2	42	1	8
2			
3	5	2	2
0			
4	15	5	5
3			

	NumStorePurchases	NumWebVisitsMonth	Complain	Response
0	4	7	0	1
1	2	5	0	0
2	10	4	0	0
3	4	6	0	0
4	6	5	0	0

[5 rows x 22 columns]

Feature engineering

```
df["Age"] = 2026-df["Year_Birth"]
```

```
df.head()
```

ID	Year_Birth	Education	Marital_Status	Income	Kidhome
Teenhome \					

0	5524	1957	Graduation	Single	58138.0	0
0						
1	2174	1954	Graduation	Single	46344.0	1
1						
2	4141	1965	Graduation	Together	71613.0	0
0						
3	6182	1984	Graduation	Together	26646.0	1
0						
4	5324	1981	PhD	Married	58293.0	1
0						

	Dt_Customer	Recency	MntWines	...	MntSweetProducts	MntGoldProds
0	04-09-2012	58	635	...	88	88
1	08-03-2014	38	11	...	1	6
2	21-08-2013	26	426	...	21	42
3	10-02-2014	26	11	...	3	5
4	19-01-2014	94	173	...	27	15

	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases
NumStorePurchases \			
0	3	8	10
4			
1	2	1	1
2			
2	1	8	2
10			
3	2	2	0
4			
4	5	5	3
6			

	NumWebVisitsMonth	Complain	Response	Age
0	7	0	1	69
1	5	0	0	72
2	4	0	0	61
3	6	0	0	42
4	5	0	0	45

[5 rows x 23 columns]

Customer Joining Date

df["Dt_Customer"] = pd.to_datetime(df["Dt_Customer"], dayfirst=True)

reference_data = df["Dt_Customer"].max()

```
df["Customer_tenure_days"] = (reference_data-
df["Dt_Customer"]).dt.days
```

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome
0	5524	1957	Graduation	Single	58138.0	0
1	2174	1954	Graduation	Single	46344.0	1
2	4141	1965	Graduation	Together	71613.0	0
3	6182	1984	Graduation	Together	26646.0	1
4	5324	1981	PhD	Married	58293.0	1

	Dt_Customer	Recency	MntWines	...	MntGoldProds	NumDealsPurchases
0	2012-09-04	58	635	...	88	3
1	2014-03-08	38	11	...	6	2
2	2013-08-21	26	426	...	42	1
3	2014-02-10	26	11	...	5	2
4	2014-01-19	94	173	...	15	5

	NumWebPurchases	NumCatalogPurchases	NumStorePurchases
0	8	10	4
1	1	1	2
2	8	2	10
3	2	0	4
4	5	3	6

	Complain	Response	Age	Customer_tenure_days
0	0	1	69	663
1	0	0	72	113
2	0	0	61	312
3	0	0	42	139
4	0	0	45	161

```
[5 rows x 24 columns]
```

```
df.columns
```

```
Index(['ID', 'Year_Birth', 'Education', 'Marital_Status', 'Income',  
      'Kidhome',  
      'Teenhome', 'Dt_Customer', 'Recency', 'MntWines', 'MntFruits',  
      'MntMeatProducts', 'MntFishProducts', 'MntSweetProducts',  
      'MntGoldProds', 'NumDealsPurchases', 'NumWebPurchases',  
      'NumCatalogPurchases', 'NumStorePurchases',  
      'NumWebVisitsMonth',  
      'Complain', 'Response', 'Age', 'Customer_tenure_days'],  
      dtype='object')
```

```
# Spending
```

```
df["Total_Spending"] = df["MntWines"]+df["MntMeatProducts"]  
+df["MntFruits"]+ df["MntFishProducts"] + df["MntSweetProducts"] +  
df["MntGoldProds"]
```

```
#Children
```

```
df["Total_Children"] = df["Kidhome"]+ df["Teenhome"]
```

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome
0	5524	1957	Graduation	Single	58138.0	0
1	2174	1954	Graduation	Single	46344.0	1
2	4141	1965	Graduation	Together	71613.0	0
3	6182	1984	Graduation	Together	26646.0	1
4	5324	1981	PhD	Married	58293.0	1

	Dt_Customer	Recency	MntWines	...	NumWebPurchases
0	2012-09-04	58	635	...	8
1	2014-03-08	38	11	...	1
2	2013-08-21	26	426	...	8
3	2014-02-10	26	11	...	2
4	2014-01-19	94	173	...	5

	NumStorePurchases	NumWebVisitsMonth	Complain	Response	Age	\
0	4	7	0	1	69	
1	2	5	0	0	72	
2	10	4	0	0	61	
3	4	6	0	0	42	
4	6	5	0	0	45	

	Customer_tenure_days	Total_Spending	Total_Children
0	663	1617	0
1	113	27	2
2	312	776	0
3	139	53	1
4	161	422	1

[5 rows x 26 columns]

Education

```
df["Education"].value_counts()
```

```
df["Education"] = df["Education"].replace({
    "Basic": "Undergraduate", "2n Cycle": "Undergraduate",
    "Graduation": "Graduate",
    "Master": "Postgraduate", "PhD": "Postgraduate"
})
```

Marital Status

Undergraduate, Graduate, Postgraduate

```
df["Living_With"] = df["Marital_Status"].replace({
    "Married": "Partner", "Together": "Partner",
    "Single": "Alone", "Divorced": "Alone",
    "Widow": "Alone", "Absurd": "Alone", "YOLO": "ALone"
})
```

```
df["Living_With"].value_counts()
```

```
Living_With
Partner    1444
Alone      794
ALone       2
Name: count, dtype: int64
```

Drop Columns

```
df.head()
```

	ID	Year_Birth	Education	Marital_Status	Income	Kidhome
Teenhome \						
0	5524	1957	Graduate	Single	58138.0	0

0						
1	2174	1954	Graduate	Single	46344.0	1
1						
2	4141	1965	Graduate	Together	71613.0	0
0						
3	6182	1984	Graduate	Together	26646.0	1
0						
4	5324	1981	Postgraduate	Married	58293.0	1
0						

	Dt_Customer	Recency	MntWines	...	NumCatalogPurchases
	NumStorePurchases	\			
0	2012-09-04	58	635	...	10
4					
1	2014-03-08	38	11	...	1
2					
2	2013-08-21	26	426	...	2
10					
3	2014-02-10	26	11	...	0
4					
4	2014-01-19	94	173	...	3
6					

	NumWebVisitsMonth	Complain	Response	Age	Customer_tenure_days	\
0	7	0	1	69	663	
1	5	0	0	72	113	
2	4	0	0	61	312	
3	6	0	0	42	139	
4	5	0	0	45	161	

	Total_Spending	Total_Children	Living_With
0	1617	0	Alone
1	27	2	Alone
2	776	0	Partner
3	53	1	Partner
4	422	1	Partner

[5 rows x 27 columns]

```
cols =
["ID","Year_Birth","Marital_Status","Kidhome","Teenhome","Dt_Customer"
]
spending_cols =
["MntWines","MntFruits","MntMeatProducts","MntFishProducts","MntSweetP
roducts","MntGoldProds"]

cols_to_drop = cols+ spending_cols

df_cleand = df.drop(columns=cols_to_drop)
```

```
df_cleand.shape
```

```
(2240, 15)
```

```
df_cleand.head()
```

	Education	Income	Recency	NumDealsPurchases	NumWebPurchases
0	Graduate	58138.0	58	3	8
1	Graduate	46344.0	38	2	1
2	Graduate	71613.0	26	1	8
3	Graduate	26646.0	26	2	2
4	Postgraduate	58293.0	94	5	5

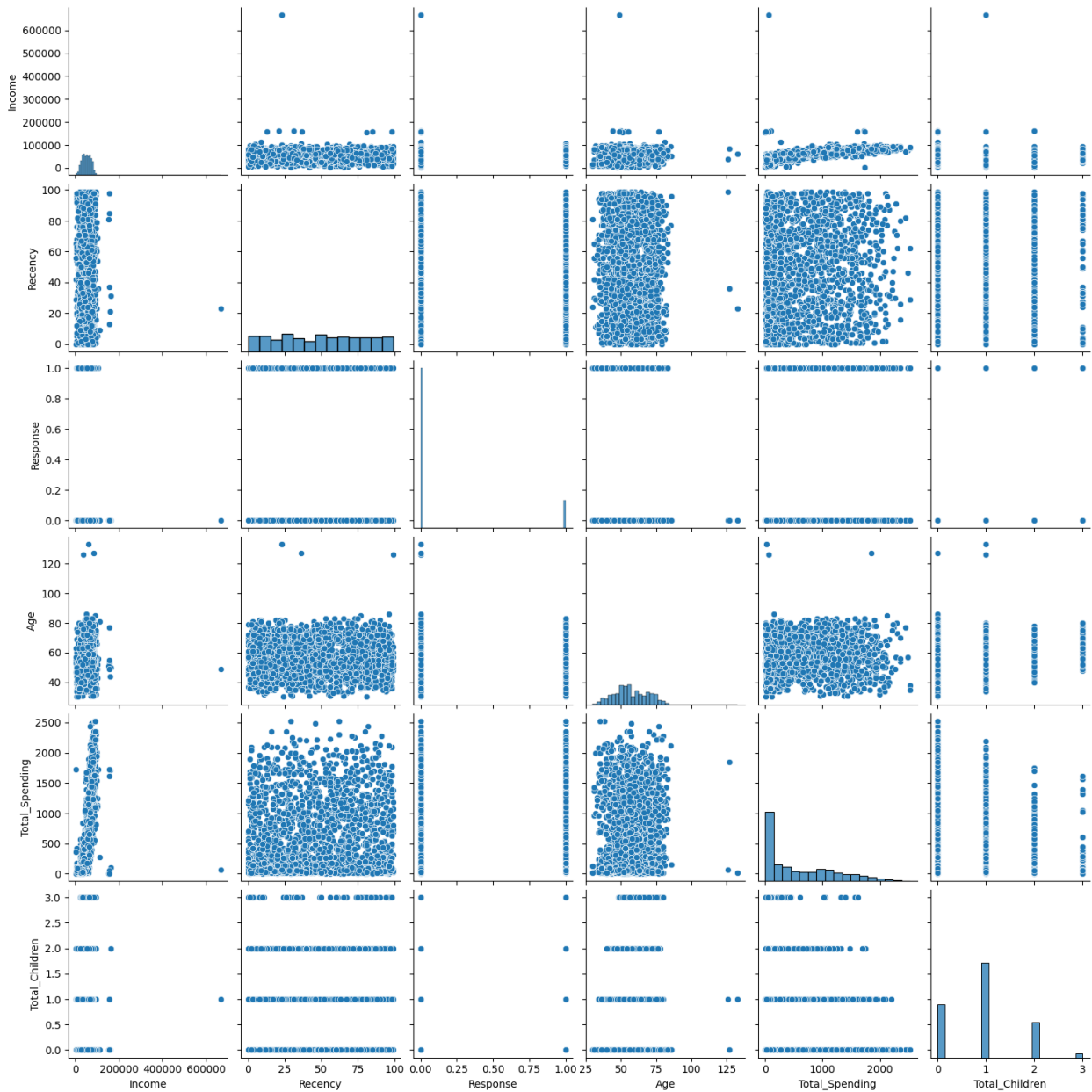
	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	Complain
0	10	4	7	0
1	1	2	5	0
2	2	10	4	0
3	0	4	6	0
4	3	6	5	0

	Response	Age	Customer_tenure_days	Total_Spending	Total_Children
0	1	69	663	1617	0
1	0	72	113	27	2
2	0	61	312	776	0
3	0	42	139	53	1
4	0	45	161	422	1

	Living_With
0	Alone
1	Alone
2	Partner
3	Partner
4	Partner

Outliers

```
cols =  
["Income", "Recency", "Response", "Age", "Total_Spending", "Total_Children"]  
  
# relative plots of some features- pair plots  
sns.pairplot(df_cleand[cols])  
  
<seaborn.axisgrid.PairGrid at 0x22b1f7a6f90>
```



```
# Remove the outliers
```

```
print("data size with outliers:", len(df_cleand))
```

```
df_cleand = df_cleand[(df_cleand["Age"]<90)]  
df_cleand = df_cleand[(df_cleand["Income"]<600_000)]
```

```
print("data size without outliers:", len(df_cleand))
```

```
data size with outliers: 2240
```

```
data size without outliers: 2236
```

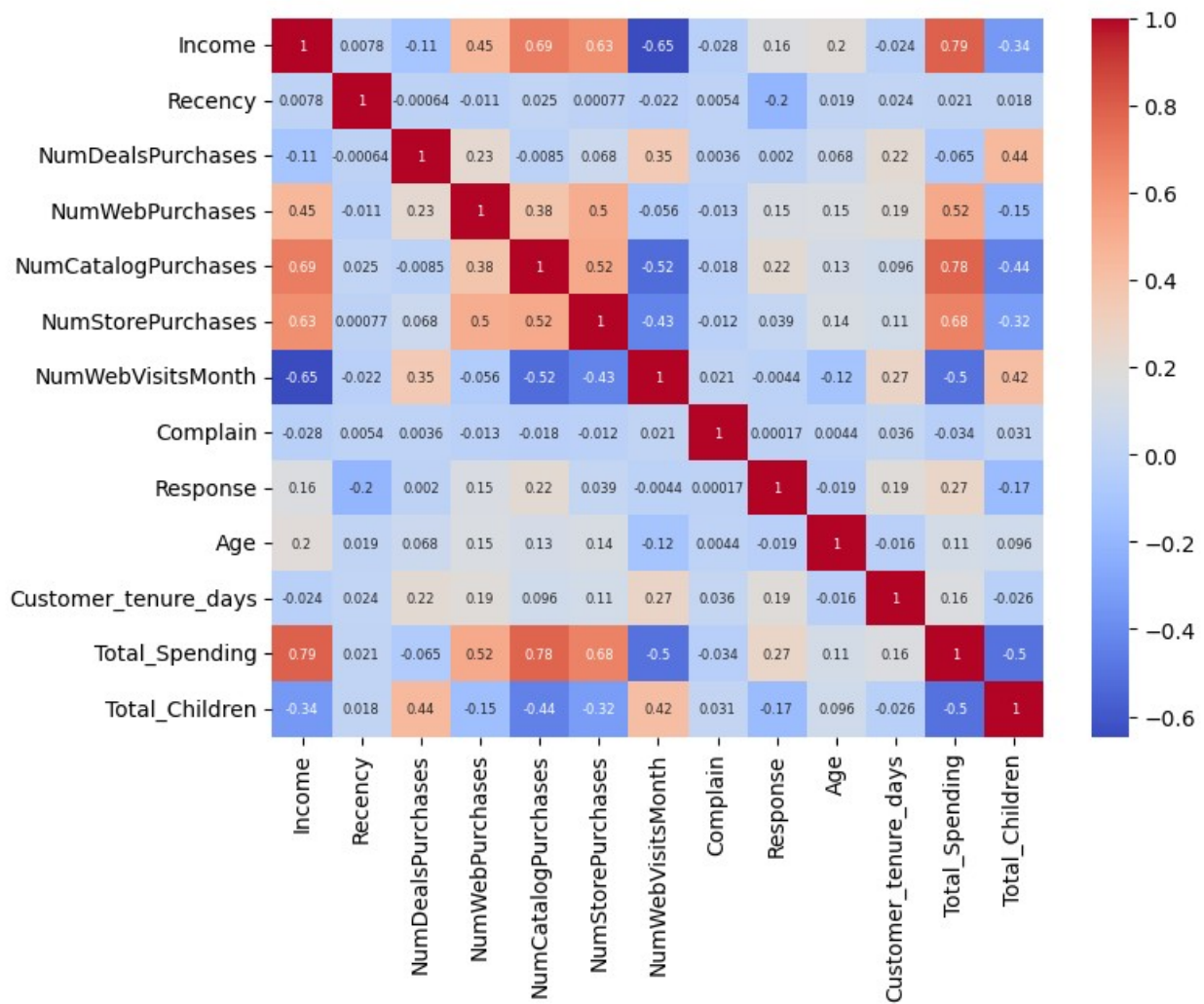
Heatmap

```
corr = df_cleand.corr(numeric_only=True)
```

```
plt.figure(figsize=(8,6))
```

```
sns.heatmap(  
    corr,  
    annot=True,  
    annot_kws={"size":6},  
    cmap="coolwarm"  
)
```

```
<Axes: >
```



```
df_cleand.shape
```

```
(2236, 15)
```

```
df_cleand.head()
```

	Education	Income	Recency	NumDealsPurchases	NumWebPurchases
0	Graduate	58138.0	58	3	8
1	Graduate	46344.0	38	2	1
2	Graduate	71613.0	26	1	8
3	Graduate	26646.0	26	2	2
4	Postgraduate	58293.0	94	5	5

	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	Complain
\				
0	10	4	7	0
1	1	2	5	0
2	2	10	4	0
3	0	4	6	0
4	3	6	5	0

	Response	Age	Customer_tenure_days	Total_Spending	Total_Children
\					
0	1	69	663	1617	0
1	0	72	113	27	2
2	0	61	312	776	0
3	0	42	139	53	1
4	0	45	161	422	1

	Living_With
0	Alone
1	Alone
2	Partner
3	Partner
4	Partner

Encoding

```

from sklearn.preprocessing import OneHotEncoder
ohe = OneHotEncoder()
cat_cols = ["Education", "Living_With"]
enc_cols = ohe.fit_transform(df_cleand[cat_cols])
enc_df =
pd.DataFrame(enc_cols.toarray(), columns=ohe.get_feature_names_out(cat_
cols), index=df_cleand.index)
df_encoded =
pd.concat([df_cleand.drop(columns=cat_cols), enc_df], axis=1)
df_encoded=df_encoded.drop(["Living_With_Alone"], axis=1)

```

```
df_encoded.shape  
(2236, 18)
```

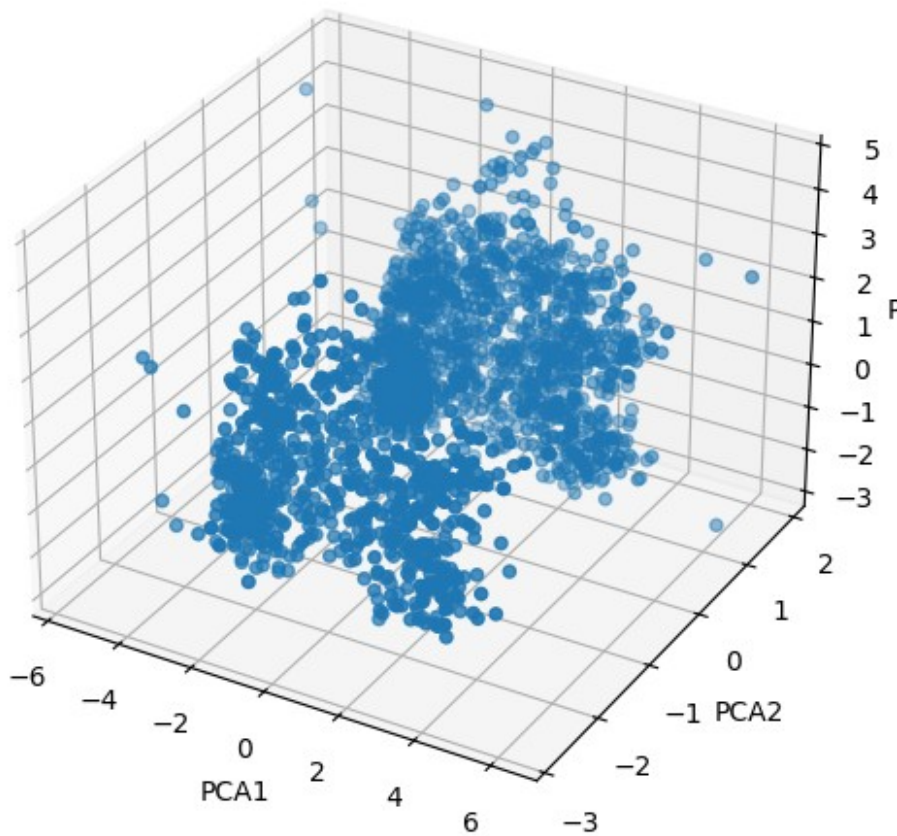
Scaling

```
from sklearn.preprocessing import StandardScaler  
X = df_encoded  
scaler = StandardScaler()  
X_scaled = scaler.fit_transform(X)
```

Visualize

```
X_scaled.shape  
(2236, 18)  
  
# 2D  
  
from sklearn.decomposition import PCA  
pca = PCA(n_components=3)  
X_pca = pca.fit_transform(X_scaled)  
pca.explained_variance_ratio_  
array([0.2316333 , 0.11373747, 0.10402889])  
  
# plot  
fig = plt.figure(figsize=(8,6))  
  
ax = fig.add_subplot(111, projection="3d")  
ax.scatter(X_pca[:, 0], X_pca[:, 1], X_pca[:, 2])  
ax.set_xlabel("PCA1")  
ax.set_ylabel("PCA2")  
ax.set_zlabel("PCA3")  
ax.set_title("3D projection")  
Text(0.5, 0.92, '3D projection')
```

3D projection



Analyze k value

1. Elbow Method

```
from sklearn.cluster import KMeans
from kneed import KneeLocator

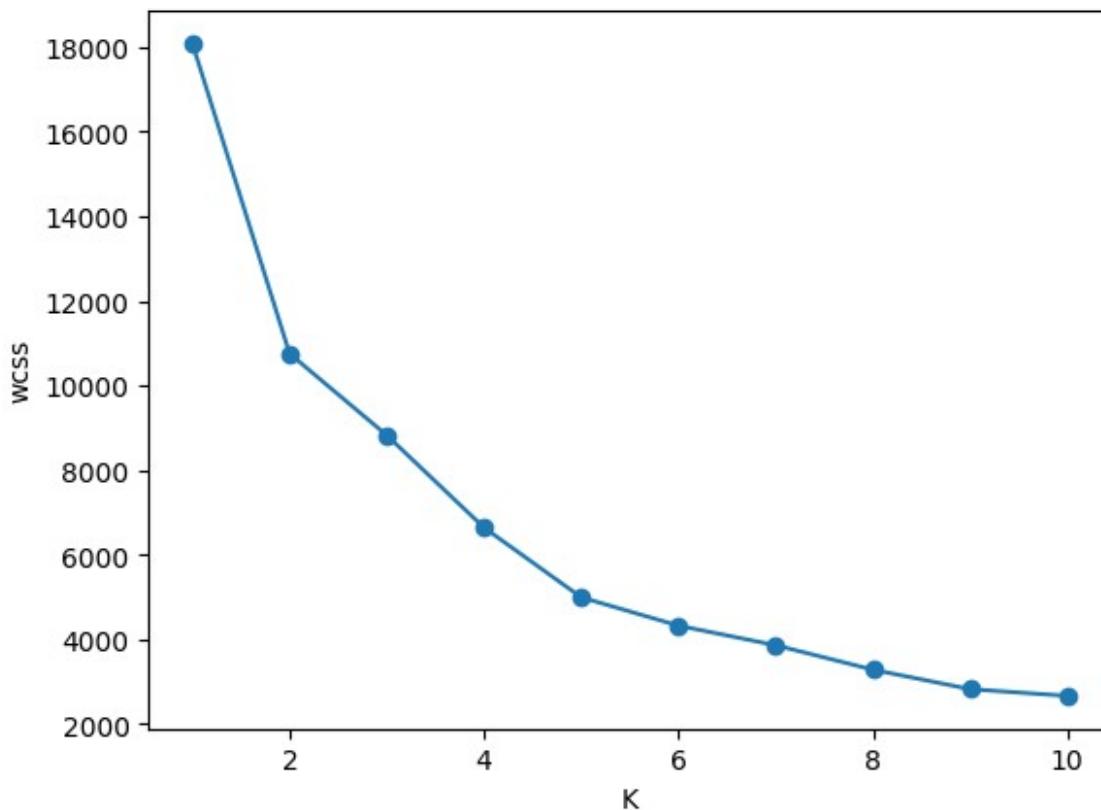
wcss = []
for k in range(1,11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit_transform(X_pca)
    wcss.append(kmeans.inertia_)

knee =
KneeLocator(range(1,11),wcss,curve="convex",direction="decreasing")
optimal_k = knee.elbow
```

```

print("best k=", optimal_k)
best k= 4
# plot
plt.plot(range(1,11), wcss,marker='o')
plt.xlabel("K")
plt.ylabel("wcss")
Text(0, 0.5, 'wcss')

```



2. Silhouette Score

```

from sklearn.metrics import silhouette_score

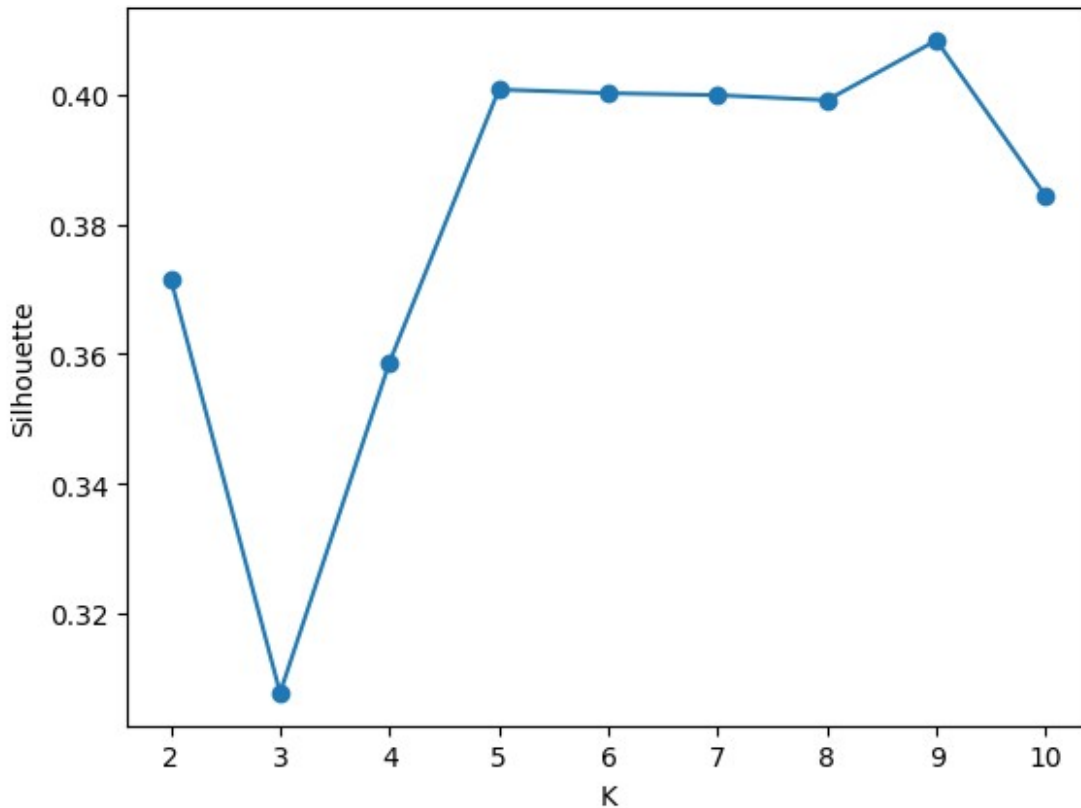
scores = []

for k in range(2,11):
    kmeans = KMeans(n_clusters=k,random_state=42)
    labels = kmeans.fit_predict(X_pca)
    score = silhouette_score(X_pca,labels)
    scores.append(score)

#plot

```

```
plt.plot(range(2,11),scores,marker = 'o')
plt.xlabel("K")
plt.ylabel("Silhouette")
Text(0, 0.5, 'Silhouette')
```



```
# combined plot

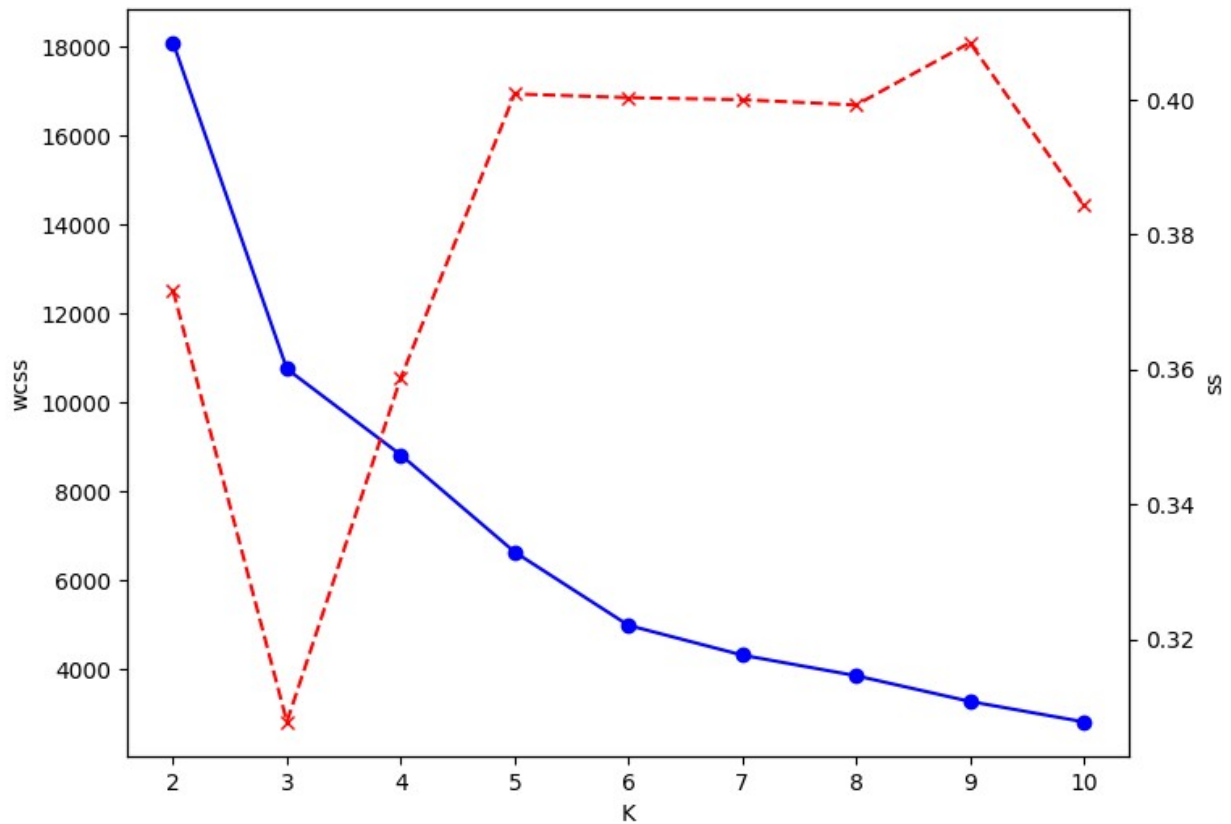
k_range = range(2, 11)

fig,ax1 = plt.subplots(figsize=(8,6))

ax1.plot(k_range,wcss[:len(k_range)], marker='o',color ="blue")
ax1.set_xlabel("K")
ax1.set_ylabel("wcss")

ax2 = ax1.twinx()
ax2.plot(k_range, scores[:len(k_range)],
marker="x",color="red",linestyle="--")
ax2.set_ylabel("ss")

Text(0, 0.5, 'ss')
```

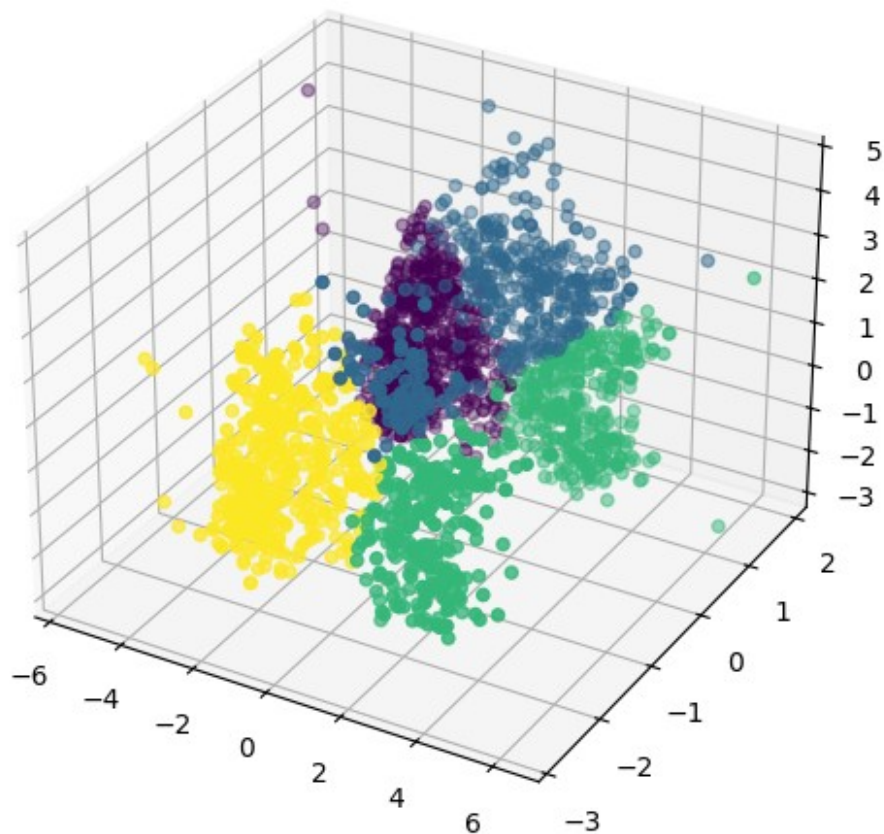


```
# k means
kmeans = KMeans(n_clusters=4, random_state=42)
labels_kmenas = kmeans.fit_predict(X_pca)

fig = plt.figure(figsize=(8,6))

ax = fig.add_subplot(111, projection="3d")
ax.scatter(X_pca[:, 0], X_pca[:, 1], X_pca[:, 2], c=labels_kmenas )

<mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x22b33afbd90>
```



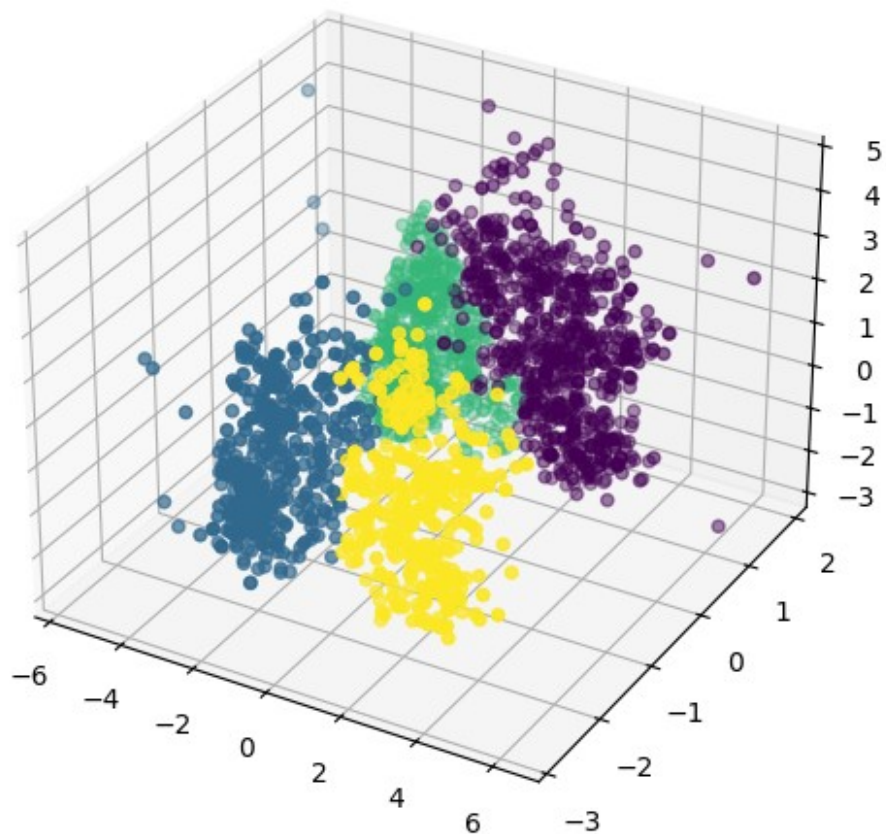
```
# Agglomerative Clustering
from sklearn.cluster import AgglomerativeClustering

agg_clf = AgglomerativeClustering(n_clusters=4, linkage="ward")
labels_agg = agg_clf.fit_predict(X_pca)

fig = plt.figure(figsize=(8,6))

ax = fig.add_subplot(111, projection="3d")
ax.scatter(X_pca[:, 0], X_pca[:, 1], X_pca[:, 2], c=labels_agg )

<mpl_toolkits.mplot3d.art3d.Path3DCollection at 0x22b33bd4cd0>
```



Characterization of Clusters

```
X["cluster"] = labels_agg
```

```
X.head()
```

	Income	Recency	NumDealsPurchases	NumWebPurchases		
NumCatalogPurchases \						
0	58138.0	58	3	8		
10						
1	46344.0	38	2	1		
1						
2	71613.0	26	1	8		
2						
3	26646.0	26	2	2		
0						
4	58293.0	94	5	5		
3						
	NumStorePurchases	NumWebVisitsMonth	Complain	Response	Age	\
0	4	7	0	1	69	

1	2	5	0	0	72
2	10	4	0	0	61
3	4	6	0	0	42
4	6	5	0	0	45

	Customer_tenure_days	Total_Spending	Total_Children
Education_Graduate \			
0	663	1617	0
1.0			
1	113	27	2
1.0			
2	312	776	0
1.0			
3	139	53	1
1.0			
4	161	422	1
0.0			

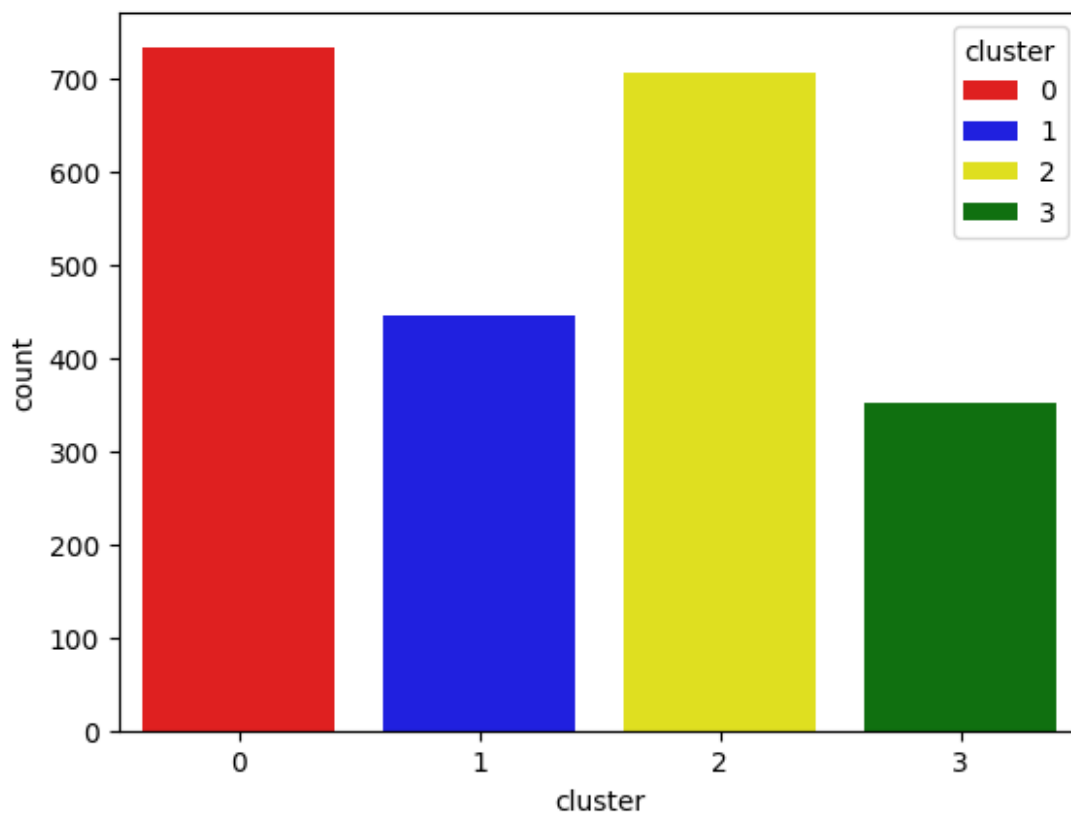
	Education_Postgraduate	Education_Undergraduate	Living_With_Alone
\			
0	0.0	0.0	1.0
1	0.0	0.0	1.0
2	0.0	0.0	0.0
3	0.0	0.0	0.0
4	1.0	0.0	0.0

	Living_With_Partner	cluster
0	0.0	3
1	0.0	1
2	1.0	0
3	1.0	2
4	1.0	0

```
pal = ["red","blue","yellow","green"]
```

```
sns.countplot(x=X["cluster"],palette=pal,hue=X["cluster"])
```

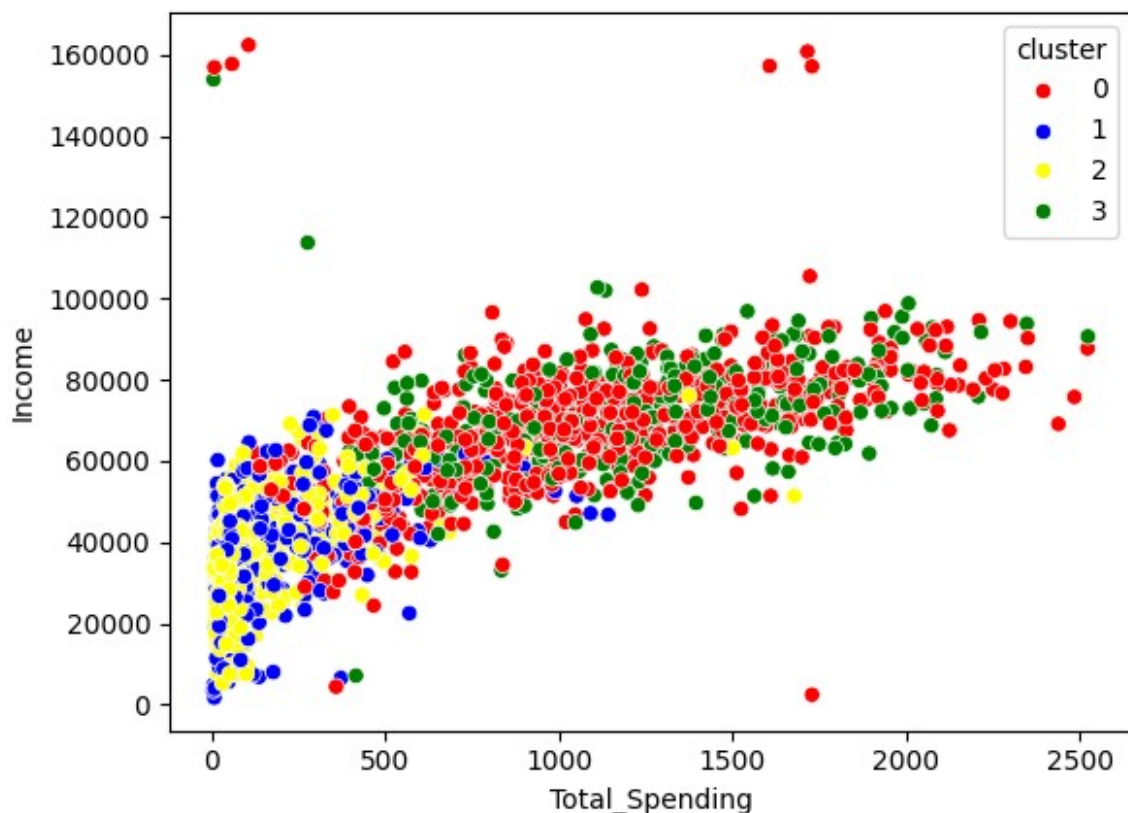
```
<Axes: xlabel='cluster', ylabel='count'>
```



Income and Spending patterns

```
sns.scatterplot(x=X["Total_Spending"],y =  
X["Income"],hue=df_cleand["cluster"],palette=pal)
```

```
<Axes: xlabel='Total_Spending', ylabel='Income'>
```



Cluster Summary

```
cluster_summary = X.groupby("cluster").mean()
print(cluster_summary)
```

	Income	Recency	NumDealsPurchases	
NumWebPurchases \ cluster				
0	67511.753070	48.691678	2.577080	5.763984
1	36995.282022	48.366292	2.595506	2.721348
2	35842.096317	49.364023	2.131728	2.359773
3	70774.174716	50.451705	1.852273	5.789773

	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth
Complain \ cluster			
0	4.601637	8.055935	4.240109
0.008186			
1	0.842697	3.633708	6.656180

0.011236			
2	0.623229	3.497167	6.390935
0.009915			
3	5.019886	8.431818	3.724432
0.005682			

	Response	Age	Customer_tenure_days	Total_Spending	\
cluster					
0	0.154161	59.792633	373.965894	1044.575716	
1	0.141573	55.710112	338.139326	166.629213	
2	0.063739	54.280453	330.983003	135.317280	
3	0.321023	58.917614	377.198864	1192.125000	

	Total_Children	Education_Graduate	Education_Postgraduate	\
cluster				
0	0.717599	0.431105	0.495225	
1	1.271910	0.487640	0.379775	
2	1.235127	0.569405	0.263456	
3	0.460227	0.542614	0.389205	

	Education_Undergraduate	Living_With_Alone
Living_With_Partner		
cluster		
0	0.073670	0.000000
1.000000		
1	0.132584	0.988764
0.006742		
2	0.167139	0.000000
1.000000		
3	0.068182	1.000000
0.000000		